Lab 7

Fuzzing for Input Validation Bugs

Hypothesis-based fuzz testing and input validation

Student Details

Name: Dhawal Pathak

SRN: PES1UG24CS151

Class: C-5

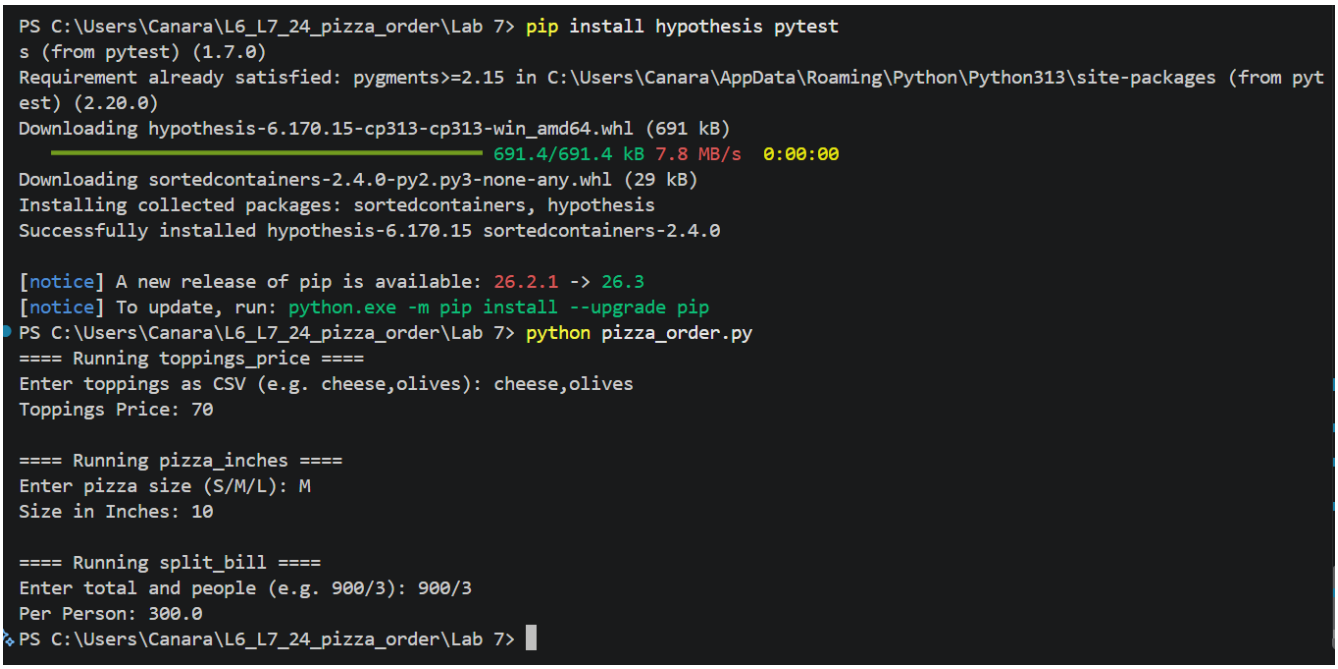
Evidence and execution screenshots

This report documents the initial application run, bugs discovered through property-based fuzzing, successful test execution after fixes, and manual checks showing that invalid inputs are handled without crashing.

Deliverable: Lab 7 screenshot report

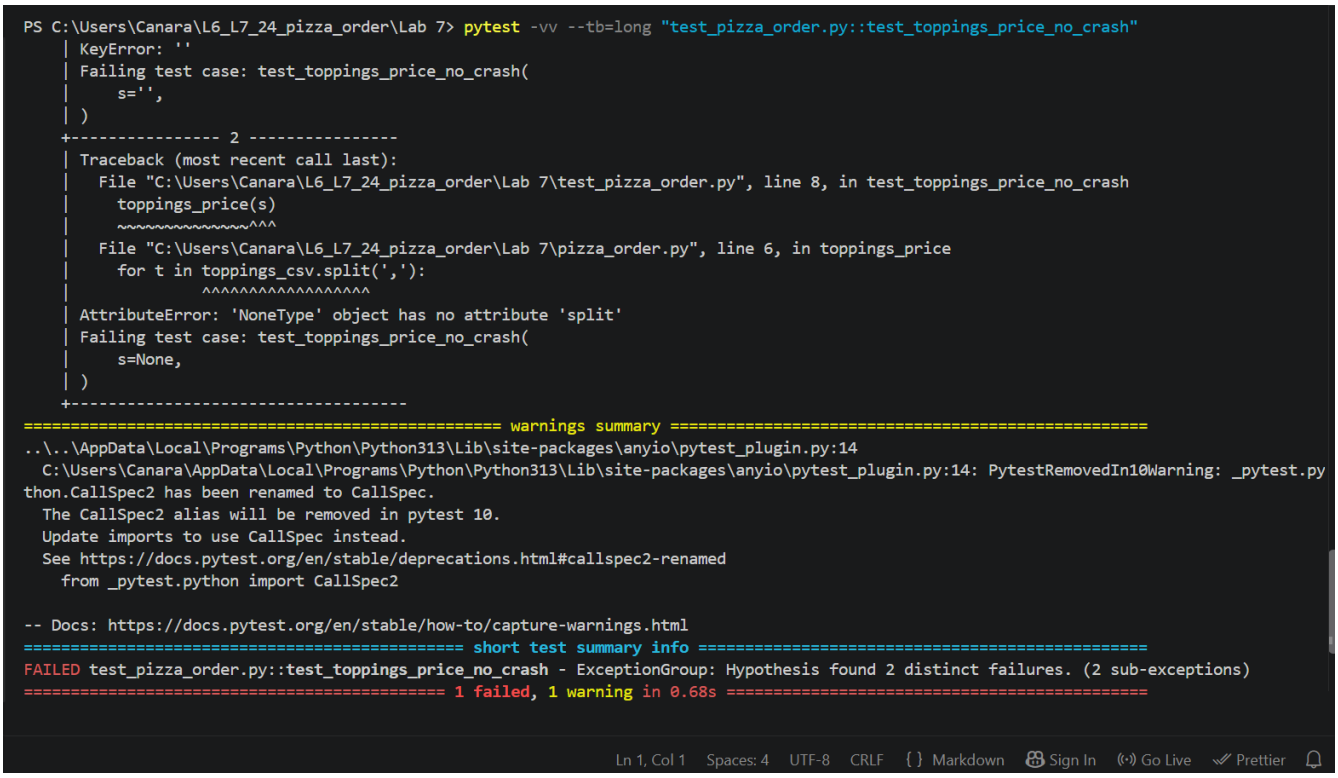
Screenshots are presented in workflow order: baseline run, fuzz-discovered failures, passing tests, and manual validation.

1. Original application



Initial execution of the provided pizza-order application using valid inputs.

2. Fuzzing: toppings price



Hypothesis discovers invalid input cases that cause the toppings_price function to fail.

3. Fuzzing: pizza size

```
PS C:\Users\Canara\L6_L7_24_pizza_order\Lab 7> pytest -vv --tb=long "test_pizza_order.py::test_pizza_inches_no_crash"
    | Failing test case: test_pizza_inches_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Canara\L6_L7_24_pizza_order\Lab 7\test_pizza_order.py", line 14, in test_pizza_inches_no_crash
    |     pizza_inches(s)
    |     ~~~~~~~~~~~~^^^
    |   File "C:\Users\Canara\L6_L7_24_pizza_order\Lab 7\pizza_order.py", line 11, in pizza_inches
    |     return SIZES[size.strip().upper()]
    |                  ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    | Failing test case: test_pizza_inches_no_crash(
    |     s=None,
    | )
    +------------------------------------
============================================================ warnings summary =============================================================
..\..\AppData\Local\Programs\Python\Python313\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\Canara\AppData\Local\Programs\Python\Python313\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.py
thon.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================================== short test summary info ===========================================================
FAILED test_pizza_order.py::test_pizza_inches_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
========================================================== 1 failed, 1 warning in 0.90s ==========================================================
PS C:\Users\Canara\L6_L7_24_pizza_order\Lab 7>
```

Hypothesis discovers invalid input cases that cause the pizza_inches function to fail.

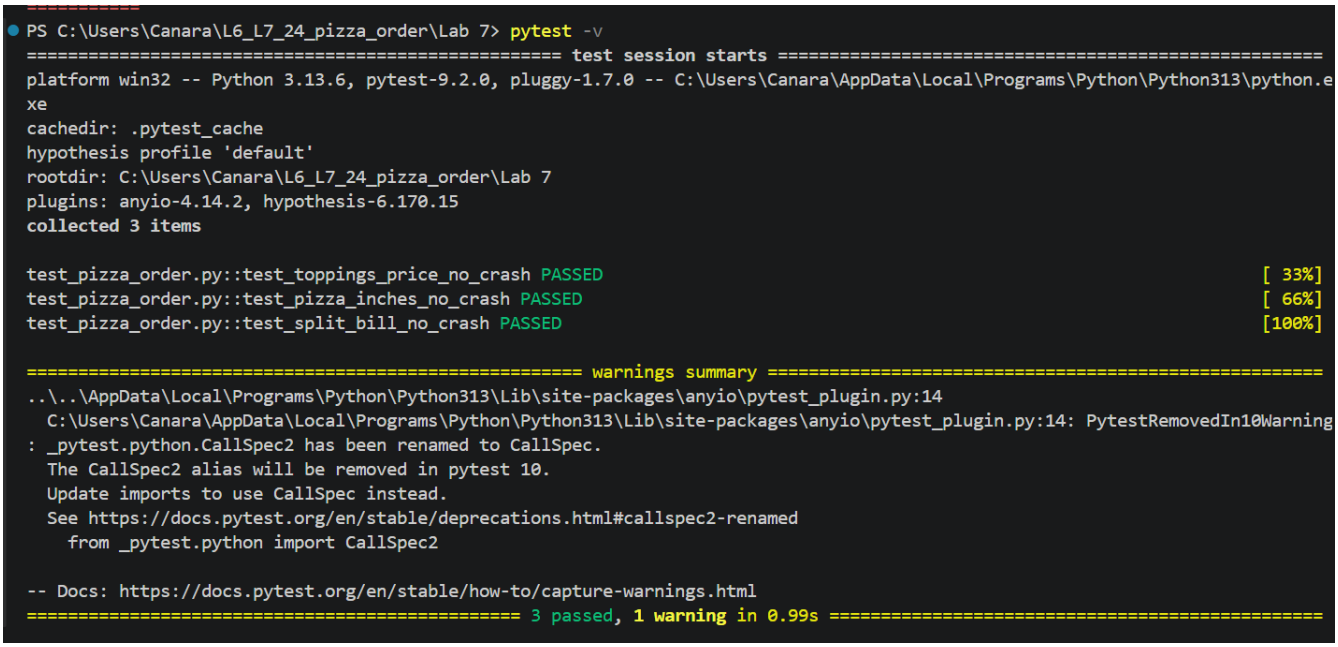
4. Fuzzing: split bill

```
PS C:\Users\Canara\L6_L7_24_pizza_order\Lab 7> pytest -vv --tb=long "test_pizza_order.py::test_split_bill_no_crash"
    | Failing test case: test_split_bill_no_crash(
    |     s='',
    | )
    +--------------- 3 ---------------
    | Traceback (most recent call last):
    |   File "C:\Users\Canara\L6_L7_24_pizza_order\Lab 7\test_pizza_order.py", line 20, in test_split_bill_no_crash
    |     split_bill(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\Canara\L6_L7_24_pizza_order\Lab 7\pizza_order.py", line 14, in split_bill
    |     total, people = text.split('/')
    |                     ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_split_bill_no_crash(
    |     s=None,
    | )
    +-----------------------------------
============================================================ warnings summary =============================================================
..\..\AppData\Local\Programs\Python\Python313\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\Canara\AppData\Local\Programs\Python\Python313\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.py
thon.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================================================== short test summary info ========================================================
FAILED test_pizza_order.py::test_split_bill_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
====================================================== 1 failed, 1 warning in 0.77s =======================================================
PS C:\Users\Canara\L6_L7_24_pizza_order\Lab 7>
```

Hypothesis discovers invalid input cases that cause the split_bill function to fail.

5. Fuzz tests passing after fixes



All three property-based fuzz tests pass after input validation fixes.

6. Invalid input validation

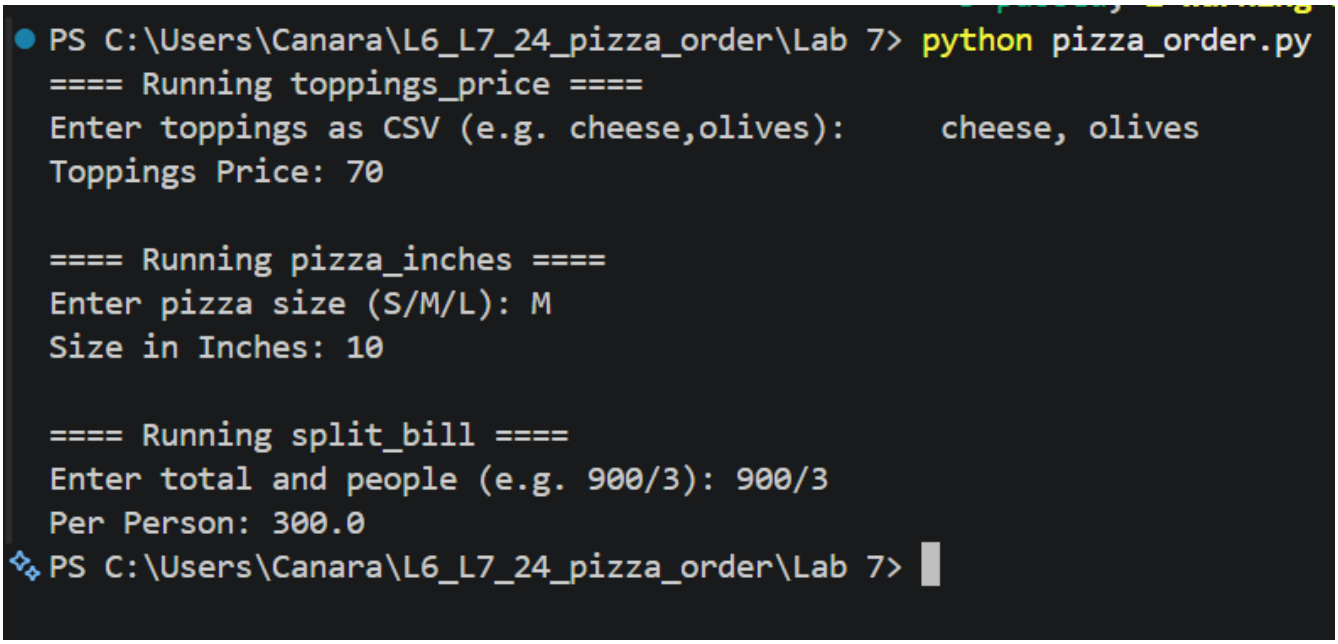
```
PS C:\Users\Canara\L6_L7_24_pizza_order\Lab 7> python pizza_order.py
==== Running toppings_price ====
Enter toppings as CSV (e.g. cheese,olives): cheese,xyz
Error: Invalid topping: xyz

==== Running pizza_inches ====
Enter pizza size (S/M/L): XXL
Error: Invalid pizza size. Choose S, M or L

==== Running split_bill ====
Enter total and people (e.g. 900/3): 900/0
Error: Number of people must be positive
```

The fixed application rejects invalid toppings, pizza size, and zero people with clear error messages.

7. Fixed application: valid inputs



The fixed application continues to work correctly with valid inputs.